

Asymptotic Inversion of the Kinkelin–Bendersky K -Function

Lambert-Anchored Transseries for the Generalized Hyperfactorial and a General Theory of $x^p \log x$ Reversion

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Abstract

The Kinkelin–Bendersky function K , characterized on the positive real axis by $K(x+1) = x^x K(x)$ and $K(1) = 1$, interpolates the hyperfactorials $K(n+1) = \prod_{k=1}^n k^k$. This article derives an all-orders asymptotic transseries for its large real inverse and for the corresponding complex sectorial inverse germs. The decisive normalization is the half-shift $t = z - \frac{1}{2}$. With A the Glaisher–Kinkelin constant and $C = \log A - \frac{1}{8}$, one has

$$\log K\left(t + \frac{1}{2}\right) \sim \frac{1}{2}t^2 \log t - \frac{1}{4}t^2 + C - \frac{1}{24} \log t + \sum_{m \geq 1} \frac{\kappa_m}{t^{2m}},$$

where

$$\kappa_m = -\frac{B_{2m+2}(1/2)}{4m(m+1)(2m+1)} = \frac{(1 - 2^{-2m-1})B_{2m+2}}{4m(m+1)(2m+1)}.$$

For $y \rightarrow +\infty$, put $Y = \log y$,

$$w = W_0(4Y/e), \quad \lambda = \frac{w+1}{2}, \quad T = e^\lambda = 2\sqrt{\frac{Y}{w}}, \quad q = T^{-2}.$$

Then the inverse branch mapping $[1, \infty)$ to $[2, \infty)$ satisfies

$$K_+^{-1}(y) = \frac{1}{2} + T + \sum_{n=1}^N a_n(\lambda) T^{1-2n} + O_N(T^{-2N-1}),$$

for every fixed N . Every a_n is an explicitly computable polynomial in λ^{-1} of degree at most $2n-1$, with coefficients in $\mathbb{Q}[C, \kappa_1, \dots, \kappa_{n-1}]$. We give a triangular recurrence, a closed ordered-operator formula obtained from a perturbative Lagrange–Bürmann identity, four explicit correction blocks, a rank-two reorganization in (q, λ^{-1}) , rigorous residual-to-error transfer, and high-precision numerical checks.

A second, accelerated Lambert anchor exactly absorbs the constant and logarithmic corrections. It gives the one-line approximation

$$K_+^{-1}(y) \approx \frac{1}{2} + \sqrt{\frac{4(Y-C)}{W_0(4(Y-C)/e)}} + \frac{1}{12},$$

whose inverse error is already $O(T^{-3}/\log T)$. Finally, the calculation is placed in a general framework for maps $F(x) = ax^p(\log x - b) + H(x)$, where H is a lower-order grid-based logarithmic transseries. The Lambert anchor is exact, the inverse support is locally finite, and the coefficients follow from a universal differential operator calculus.

Keywords. Kinkelin–Bendersky K -function; hyperfactorial; generalized hyperfactorial; Glaisher–Kinkelin constant; Barnes G -function; Lambert W ; asymptotic inverse; transseries; Lagrange–Bürmann inversion; series reversion.

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1 The inverse problem and the answer in compact form

The object of study is the real-valued Kinkelin–Bendersky function $K : [0, \infty) \rightarrow (0, \infty)$, normalized by

$$K(0) = K(1) = K(2) = 1, \quad K(x+1) = x^x K(x) \quad (x > 0). \quad (1)$$

At positive integers,

$$K(n+1) = \prod_{k=1}^n k^k. \quad (2)$$

This is the convention used in the standard literature on the K -function [2, 7]. Some software and combinatorial sources instead denote by $H(x)$ the shifted hyperfactorial

$$H(x) := K(x+1). \quad (3)$$

Thus an inverse formula for H is obtained from the formulas below by subtracting 1 from the inverse of K .

For large positive arguments, the relevant inverse is the increasing branch

$$K_+^{-1} : [1, \infty) \rightarrow [2, \infty). \quad (4)$$

The endpoint value $y = 1$ is not globally one-to-one because $K(0) = K(1) = K(2) = 1$; the subscript $+$ fixes the branch which grows to infinity.

Main result

Let

$$Y = \log y, \quad w = W_0\left(\frac{4Y}{e}\right), \quad \lambda = \frac{w+1}{2}, \quad T = e^\lambda = 2\sqrt{\frac{Y}{w}}. \quad (5)$$

Let A be the Glaisher–Kinkelin constant and set

$$C = \log A - \frac{1}{8} = -\zeta'(-1) - \frac{1}{24}. \quad (6)$$

Then, as $y \rightarrow +\infty$,

$$K_+^{-1}(y) \sim \frac{1}{2} + T + \sum_{n \geq 1} a_n(\lambda) T^{1-2n}. \quad (7)$$

The first three coefficients are

$$a_1(\lambda) = \frac{1}{24} - \frac{C}{\lambda}, \quad (8)$$

$$a_2(\lambda) = -\frac{1}{1152} + \frac{20C+1}{480\lambda} - \frac{C^2}{2\lambda^2} - \frac{C^2}{2\lambda^3}, \quad (9)$$

$$a_3(\lambda) = \frac{1}{27648} - \frac{1890C+311}{725760\lambda} + \frac{C(10C+1)}{160\lambda^2} - \frac{C(240C^2-10C-1)}{480\lambda^3} - \frac{5C^3}{6\lambda^4} - \frac{C^3}{2\lambda^5}. \quad (10)$$

For every fixed $N \geq 0$, truncation after a_N has the real-axis error

$$K_+^{-1}(y) - \left(\frac{1}{2} + T + \sum_{n=1}^N a_n(\lambda) T^{1-2n} \right) = O_N(T^{-2N-1}). \quad (11)$$

The convention is that the sum is empty when $N = 0$.

The first correction can be written directly in Y and w :

$$K_+^{-1}(y) = \frac{1}{2} + 2\sqrt{\frac{Y}{w}} + \frac{1}{2}\sqrt{\frac{w}{Y}} \left(\frac{1}{24} - \frac{2C}{w+1} \right) + O\left(\left(\frac{w}{Y}\right)^{3/2}\right). \quad (12)$$

This already displays the two essential scales. The Lambert variable $w \sim \log Y$ resums the entire iterated-logarithm expansion of the leading block. The first genuinely new block is smaller by

$$T^{-2} = e^{-(w+1)} = \frac{w}{4Y}, \quad (13)$$

which is beyond every algebraic order in $1/w$.

1.1 What is meant by a transseries here

There are three equivalent ways to read [Equation \(7\)](#).

1. As a *Lambert-anchored outer expansion*, it is a Poincaré expansion in powers of $q = T^{-2}$, with coefficient functions of the slow variable $\lambda = \log T$.
2. As a *rank-two grid-based transseries*, its monomials are

$$Tq^n \lambda^{-j} = \exp\left(-\frac{2n-1}{2}(w+1)\right) \lambda^{-j}, \quad n \geq 1, \quad (14)$$

with $0 \leq j \leq 2n-1$ in the present problem.

3. After expanding $w = W_0(4Y/e)$ in $L_1 = \log(4Y/e)$ and $L_2 = \log L_1$, it becomes a nested logarithmic transseries in powers of L_1/Y , whose coefficient blocks are polynomials in L_2 and inverse powers of L_1 .

The Lambert-anchored form is both shorter and numerically superior. Expanding W too early destroys an exact inversion and replaces it with an infinite slow series before the first algebraic correction has even appeared.

1.2 Outline and status of the results

The forward asymptotic expansion of the hyperfactorial and of related Bendersky functions is classical and has been refined in a substantial literature; see, for example, [2, 10, 9, 11]. A useful closed Lambert approximation to the inverse has also appeared in an online discussion [6]. The all-orders inverse coefficients, the centered operator formula, the rank-two resummation, and the general $ax^p(\log x - b)$ inversion calculus are derived here. This statement is a description of the contents of this note, not a claim that no equivalent formula can exist elsewhere.

The argument deliberately separates three levels of validity.

- The identities for coefficients are exact formal identities in a locally finite transseries algebra.
- On the positive real axis, the source expansion and monotonicity turn each fixed truncation into the rigorous asymptotic estimate [Equation \(11\)](#).
- In the complex plane, the same formulas describe local inverse germs on admissible sectors, once the logarithm and Lambert branches are fixed and the path stays away from critical points.

2 The K -function and its increasing real branch

2.1 Integral, zeta, and Barnes representations

For $x \geq 0$, one convenient definition is

$$\log K(x) = \frac{x^2 - x}{2} - \frac{x}{2} \log(2\pi) + \int_0^x \log \Gamma(u) \, du. \quad (15)$$

The singularity of $\log \Gamma(u)$ at the origin is integrable. A standard complex continuation is encoded by

$$K(z) = \exp(\zeta'(-1, z) - \zeta'(-1)), \quad (16)$$

with compatible branch conventions. On the positive real axis, the relation to the Barnes G -function is

$$K(z)G(z) = \Gamma(z)^{z-1}. \quad (17)$$

Equivalently,

$$K(x+1)G(x+1) = \Gamma(x+1)^x. \quad (18)$$

These formulas and the recurrence [Equation \(1\)](#) are recorded in [7]; the Barnes asymptotics used below are summarized in the NIST Digital Library of Mathematical Functions [5].

2.2 Monotonicity beyond 2

Differentiate [Equation \(15\)](#):

$$\frac{d}{dx} \log K(x) = x - \frac{1}{2} - \frac{1}{2} \log(2\pi) + \log \Gamma(x), \quad (19)$$

so that

$$\frac{d^2}{dx^2} \log K(x) = 1 + \psi(x), \quad (20)$$

where $\psi = \Gamma'/\Gamma$.

Proposition 2.1 (Increasing real branch). *The function K is strictly increasing on $[2, \infty)$, maps this interval onto $[1, \infty)$, and therefore has the inverse branch [Equation \(4\)](#).*

Proof. The digamma function is increasing on $(0, \infty)$, and $\psi(2) = 1 - \gamma > 0$. Hence $1 + \psi(x) > 0$ for $x \geq 2$, so the right-hand side of [Equation \(19\)](#) is increasing there. At $x = 2$, it equals

$$\frac{3}{2} - \frac{1}{2} \log(2\pi) > 0.$$

Thus $(\log K)'(x) > 0$ for $x \geq 2$, and therefore $K'(x) > 0$. Since $K(2) = 1$ and the large- x expansion in [Section 3](#) tends to infinity, the range is $[1, \infty)$. $\square$

Corollary 2.2 (Shifted hyperfactorial inverse). *If $H(x) = K(x+1)$, then on the increasing branch*

$$H_+^{-1}(y) = K_+^{-1}(y) - 1 \sim -\frac{1}{2} + T + \sum_{n \geq 1} a_n(\lambda) T^{1-2n}. \quad (21)$$

3 The centered forward expansion

The forward asymptotic expansion is not merely an input to inversion. Its correct centering determines the support and the complexity of the inverse transseries.

3.1 The classical uncentered form

The Barnes expansion [5, §5.17(ii)] and the exact cancellation in [Equation \(18\)](#) give, uniformly in every closed sector $|\arg z| \leq \pi - \delta$,

$$\begin{aligned} \log K(z+1) &\sim \left(\frac{z^2}{2} + \frac{z}{2} + \frac{1}{12} \right) \log z - \frac{z^2}{4} + \log A \\ &\quad - \sum_{m \geq 1} \frac{B_{2m+2}}{4m(m+1)(2m+1)z^{2m}}. \end{aligned} \quad (22)$$

Here B_n denotes the Bernoulli number and

$$\log A = \frac{1}{12} - \zeta'(-1). \quad (23)$$

Although Equation (22) contains only even negative powers, its leading polynomial-logarithmic part contains a linear term $z \log z$. The inverse becomes substantially simpler after the half-shift.

3.2 Half-shift and parity

Define

$$t = z - \frac{1}{2}, \quad \mathcal{L}(t) := \log K\left(t + \frac{1}{2}\right), \quad \Phi(t) := \frac{1}{2}t^2 \log t - \frac{1}{4}t^2. \quad (24)$$

Let the Bernoulli polynomials be normalized by

$$\frac{ue^{xu}}{e^u - 1} = \sum_{n \geq 0} B_n(x) \frac{u^n}{n!}. \quad (25)$$

They satisfy

$$B_n\left(\frac{1}{2}\right) = (2^{1-n} - 1)B_n. \quad (26)$$

Theorem 3.1 (Canonical centered expansion). *As $t \rightarrow \infty$ in every closed sector $|\arg t| \leq \pi - \delta$,*

$$\mathcal{L}(t) \sim \Phi(t) + C - \frac{1}{24} \log t + \sum_{m \geq 1} \kappa_m t^{-2m}, \quad (27)$$

where C is given by Equation (6) and

$$\kappa_m = -\frac{B_{2m+2}(1/2)}{4m(m+1)(2m+1)} = \frac{(1 - 2^{-2m-1})B_{2m+2}}{4m(m+1)(2m+1)}. \quad (28)$$

For every fixed $M \geq 1$, the corresponding finite form is

$$\mathcal{L}(t) = \Phi(t) + C - \frac{1}{24} \log t + \sum_{m=1}^{M-1} \kappa_m t^{-2m} + O(t^{-2M}). \quad (29)$$

Proof. From Equation (19), with $x = t + \frac{1}{2}$,

$$\mathcal{L}'(t) = t - \frac{1}{2} \log(2\pi) + \log \Gamma\left(t + \frac{1}{2}\right). \quad (30)$$

The shifted Stirling expansion is

$$\log \Gamma(t + a) \sim \left(t + a - \frac{1}{2}\right) \log t - t + \frac{1}{2} \log(2\pi) + \sum_{n \geq 1} \frac{(-1)^{n+1} B_{n+1}(a)}{n(n+1)t^n}. \quad (31)$$

At $a = \frac{1}{2}$, all Bernoulli polynomials of odd degree greater than one vanish. Hence

$$\mathcal{L}'(t) \sim t \log t - \frac{1}{24t} + \sum_{m \geq 1} \frac{B_{2m+2}(1/2)}{(2m+1)(2m+2)t^{2m+1}}. \quad (32)$$

Termwise integration gives Equation (27) up to a constant and yields exactly Equation (28). Matching with the uncentered expansion Equation (22) determines the constant as $\log A - \frac{1}{8} = C$. Uniform remainder estimates follow either by applying the standard sectorial remainder for shifted Stirling and integrating, or directly by re-expanding the Barnes remainder. $\square$

The first coefficients are

$$\kappa_1 = -\frac{7}{5760}, \quad \kappa_2 = \frac{31}{161280}, \quad \kappa_3 = -\frac{127}{1290240}, \quad \kappa_4 = \frac{511}{4866048}, \quad \kappa_5 = -\frac{1414477}{7380172800}. \quad (33)$$

Remark 3.2 (Why the half-shift is canonical). The half-shift simultaneously removes the $z \log z$ term from the dominant phase and converts the entire tail to even inverse powers. At the derivative level this is the identity $B_{2m+1}(1/2) = 0$. Consequently the inverse corrections occur in the sparse lattice T^{1-2n} , rather than in every integer power of T^{-1} .

Scale and branch warning

The variable in Equation (27) is $t = z - \frac{1}{2}$, where z is the argument of K . If one instead uses the common hyperfactorial variable x in $H(x) = K(x+1)$, then $t = x + \frac{1}{2}$. Losing this shift produces an apparent error of order $1/2$, much larger than every displayed correction block.

4 Exact Lambert inversion of the dominant phase

The leading phase is

$$\Phi(t) = \frac{1}{4}t^2(\log(t^2) - 1). \quad (34)$$

It is not necessary to invert this phase asymptotically: it is exactly Lambert-invertible.

Proposition 4.1 (Exact dominant anchor). *Let $Y > 0$ and define w, λ, T by Equation (5). Then*

$$\Phi(T) = Y, \quad \log T = \lambda, \quad T^2 = \frac{4Y}{w}, \quad T^{-2} = e^{-(w+1)} = \frac{w}{4Y}. \quad (35)$$

Proof. Set $w = 2 \log T - 1$. Then

$$4\Phi(T) = T^2(2 \log T - 1) = e^{w+1}w. \quad (36)$$

Thus $\Phi(T) = Y$ is equivalent to

$$we^w = \frac{4Y}{e}. \quad (37)$$

On the positive real axis this gives $w = W_0(4Y/e)$, followed by all the identities in Equation (35). $\square$

Corollary 4.2 (Leading inverse). *As $y \rightarrow +\infty$,*

$$K_+^{-1}(y) = \frac{1}{2} + T + O(T^{-1}). \quad (38)$$

Proof. The omitted part of Equation (27) is $O(\log T)$, while $\Phi'(T) = T \log T$. One inverse-function step therefore produces a displacement $O(T^{-1})$. $\square$

The two slow-fast parameters used throughout the rest of the article are

$$\lambda = \log T = \frac{w+1}{2}, \quad q = T^{-2} = e^{-2\lambda}. \quad (39)$$

They satisfy the strict scale separation

$$q = o(\lambda^{-M}) \quad (M = 0, 1, 2, \dots). \quad (40)$$

Thus a correction of relative size q is beyond all orders of the ordinary large- λ expansion.

5 The normalized inverse equation and triangular reversion

Write the unknown centered inverse in the form

$$t = T(1 + u), \quad u \sim \sum_{n \geq 1} a_n(\lambda) q^n, \quad q = T^{-2}. \quad (41)$$

Because $\Phi(T) = Y$, direct substitution gives

$$\frac{\Phi(T(1 + u)) - \Phi(T)}{T^2} = \Psi_\lambda(u), \quad (42)$$

where

$$\Psi_\lambda(u) := \frac{1}{2}(1 + u)^2 \left(\lambda + \log(1 + u) - \frac{1}{2} \right) - \frac{1}{2} \left(\lambda - \frac{1}{2} \right). \quad (43)$$

In particular,

$$\Psi_\lambda(0) = 0, \quad \partial_u \Psi_\lambda(0) = \lambda. \quad (44)$$

Let

$$\mathcal{H}(q, \lambda) := C - \frac{\lambda}{24} + \sum_{m \geq 1} \kappa_m q^m. \quad (45)$$

The complete formal inverse equation is

$$\begin{aligned} 0 &= \mathcal{F}(u, q; \lambda) \\ &:= \Psi_\lambda(u) + q \left(C - \frac{\lambda + \log(1 + u)}{24} \right) + \sum_{m \geq 1} \kappa_m q^{m+1} (1 + u)^{-2m}. \end{aligned} \quad (46)$$

The equality is to be read formally when the full divergent source series is used, and as an asymptotic identity to any prescribed finite order when [Equation \(29\)](#) is truncated.

Theorem 5.1 (Unique formal inverse and triangular recurrence). *There is a unique formal solution*

$$U(q; \lambda) = \sum_{n \geq 1} a_n(\lambda) q^n \quad (47)$$

to $\mathcal{F}(U, q; \lambda) = 0$ with $U(0; \lambda) = 0$, after adjoining λ^{-1} to the coefficient ring. If

$$U_{N-1}(q; \lambda) = \sum_{j=1}^{N-1} a_j(\lambda) q^j, \quad (48)$$

then

$$a_N(\lambda) = -\frac{1}{\lambda} [q^N] \mathcal{F}(U_{N-1}(q; \lambda), q; \lambda). \quad (49)$$

Only $\kappa_1, \dots, \kappa_{N-1}$ are needed to compute a_N .

Proof. Suppose $a_1, \dots, a_{N-1}$ have already been determined. Replace U_{N-1} by $U_{N-1} + a_N q^N$ in [Equation \(46\)](#). At order q^N , the new unknown can enter only through the linearization of Ψ_λ at $u = 0$; all nonlinear occurrences multiply it by an additional positive power of q . By [Equation \(44\)](#), its coefficient is λa_N . Setting the coefficient of q^N to zero gives [Equation \(49\)](#). This also proves uniqueness by induction. The source term $\kappa_m q^{m+1}$ can affect order q^N only when $m \leq N - 1$. $\square$

The recurrence is often the fastest way to generate coefficients in a computer algebra system. It also reveals why the expansion is locally finite: no coefficient requires an infinite part of the forward asymptotic series.

5.1 A polynomial structure theorem

It is useful to isolate the dependence on the second small parameter

$$s = \lambda^{-1}. \quad (50)$$

Decompose

$$\Psi_\lambda(u) = \lambda A(u) + B(u), \quad (51)$$

where

$$A(u) = u + \frac{u^2}{2}, \quad (52)$$

$$B(u) = \frac{1}{2}(1+u)^2 \log(1+u) - \frac{u}{2} - \frac{u^2}{4}. \quad (53)$$

After division by λ , the implicit equation becomes

$$A(u) - \frac{q}{24} + sR(u, q) = 0, \quad (54)$$

with

$$R(u, q) = B(u) + q \left(C - \frac{\log(1+u)}{24} \right) + \sum_{m \geq 1} \kappa_m q^{m+1} (1+u)^{-2m}. \quad (55)$$

All explicit dependence on λ has now been compressed into one factor s .

Theorem 5.2 (Finite slow depth of each outer block). *For every $N \geq 1$,*

$$a_N(\lambda) \in \mathbb{Q}[C, \kappa_1, \dots, \kappa_{N-1}][\lambda^{-1}], \quad \deg_{\lambda^{-1}} a_N \leq 2N - 1. \quad (56)$$

Moreover,

$$[\lambda^0]a_N(\lambda) = \binom{1/2}{N} 12^{-N}. \quad (57)$$

Proof. At $s = 0$, Equation (54) reduces to

$$u + \frac{u^2}{2} = \frac{q}{24}. \quad (58)$$

The solution with $u(0) = 0$ is

$$u = \sqrt{1 + \frac{q}{12}} - 1, \quad (59)$$

which proves Equation (57). For the degree bound, use induction on N in the coefficient equation obtained from Equation (54). The term linear in a_N has coefficient one. A product of previously known coefficients with total outer degree N has slow degree at most $2N - 2$; multiplication by the single explicit factor s in sR raises this to at most $2N - 1$. Formal composition with $\log(1+u)$ and $(1+u)^{-2m}$ preserves the same outer-degree estimate. The coefficient field is generated only by the source constants that can occur through order q^N . $\square$

Remark 5.3. The finiteness in Equation (56) is special and stronger than what is needed for a transseries. A general coefficient differential algebra may produce an infinite slow expansion inside each outer block. Here the linear appearance of λ in the centered source equation forces a polynomial in λ^{-1} .

6 Perturbative Lagrange–Bürmann inversion

The triangular recurrence is constructive, but an all-orders closed formula is more revealing. The needed identity is a perturbative form of Lagrange–Bürmann inversion; related uses of Lagrange inversion in asymptotic factorial problems appear in [3].

6.1 A general differential-operator identity

Theorem 6.1 (Perturbative Lagrange–Bürmann formula). *Let F_0 and H be analytic near X , suppose $F_0'(X) \neq 0$, and let $x(\varepsilon)$ be the local solution of*

$$F_0(x) + \varepsilon H(x) = F_0(X), \quad x(0) = X. \quad (60)$$

Define

$$\mathcal{D}_0 := \frac{1}{F_0'(X)} \frac{d}{dX}. \quad (61)$$

Then, as a convergent local power series in ε , and also as a formal identity,

$$x(\varepsilon) = X + \sum_{n \geq 1} \frac{(-\varepsilon)^n}{n!} \mathcal{D}_0^{n-1} \left(\frac{H(X)^n}{F_0'(X)} \right). \quad (62)$$

More generally, for analytic g ,

$$g(x(\varepsilon)) = g(X) + \sum_{n \geq 1} \frac{(-\varepsilon)^n}{n!} \mathcal{D}_0^{n-1} \left(\frac{g'(X)H(X)^n}{F_0'(X)} \right). \quad (63)$$

Proof. Take a small positively oriented contour around X containing no other zero of $F_0(z) - F_0(X)$. For sufficiently small ε , the argument principle gives

$$g(x(\varepsilon)) = \frac{1}{2\pi i} \oint g(z) \frac{F_0'(z) + \varepsilon H'(z)}{F_0(z) + \varepsilon H(z) - F_0(X)} dz. \quad (64)$$

Subtract the same formula at $\varepsilon = 0$, write the difference as a logarithmic derivative, integrate by parts, and expand $\log(1 + \varepsilon H/(F_0 - F_0(X)))$. The coefficient of ε^n is

$$\frac{(-1)^n}{n} \operatorname{Res}_{z=X} \frac{g'(z)H(z)^n}{(F_0(z) - F_0(X))^n}. \quad (65)$$

The change of local coordinate $v = F_0(z)$ yields the standard residue identity

$$\operatorname{Res}_{z=X} \frac{\varphi(z)}{(F_0(z) - F_0(X))^n} = \frac{1}{(n-1)!} \left(\frac{1}{F_0'(z)} \frac{d}{dz} \right)^{n-1} \frac{\varphi(z)}{F_0'(z)} \Big|_{z=X}. \quad (66)$$

Combining the two formulas proves Equation (63); taking $g(z) = z$ gives Equation (62). $\square$

Reusable method

To invert $F = F_0 + H$, first solve $F_0(X) = Y$ exactly or asymptotically, then apply Equation (62) with $\varepsilon = 1$. If the operator $\mathcal{D}_0$ strictly raises a chosen asymptotic valuation, the infinite sum is locally finite coefficient by coefficient and is valid as a formal transseries even when it is not a convergent perturbation series.

6.2 Specialization to the K -function

For the centered K -problem,

$$F_0(T) = \Phi(T), \quad F_0'(T) = T\lambda, \quad H(T) = C - \frac{\lambda}{24} + \sum_{m \geq 1} \kappa_m T^{-2m}. \quad (67)$$

Therefore

$$\mathcal{D} := \frac{1}{T\lambda} \frac{d}{dT} \quad (68)$$

and the centered inverse has the formal expansion

$$t \sim T + \sum_{n \geq 1} \frac{(-1)^n}{n!} \mathcal{D}^{n-1} \left(\frac{H(T)^n}{T\lambda} \right). \quad (69)$$

Equation (69) is an all-orders formula requiring no implicit coefficient solving.

To make the outer blocks explicit, observe that

$$\mathcal{D}(Tq^r f(\lambda)) = Tq^{r+1} \mathcal{A}_r f(\lambda), \quad (70)$$

where

$$\mathcal{A}_r f := \frac{f'(\lambda) + (1 - 2r)f(\lambda)}{\lambda}. \quad (71)$$

Here and below, a prime on a coefficient function means differentiation with respect to λ .

Theorem 6.2 (Ordered block formula). *Let*

$$\mathcal{H}(q, \lambda) = C - \frac{\lambda}{24} + \sum_{m \geq 1} \kappa_m q^m. \quad (72)$$

Then the coefficient of $Tq^N = T^{1-2N}$ in the inverse is

$$a_N(\lambda) = \sum_{n=1}^N \frac{(-1)^n}{n!} (\mathcal{A}_{N-1} \circ \mathcal{A}_{N-2} \circ \cdots \circ \mathcal{A}_{N-n+1}) \times \left(\frac{[q^{N-n}] \mathcal{H}(q, \lambda)^n}{\lambda} \right). \quad (73)$$

For $n = 1$, the displayed operator composition is empty and acts as the identity. In every nonempty composition, the rightmost operator acts first.

Proof. In the n -th summand of Equation (69), a term $[q^m] \mathcal{H}^n$ begins as Tq^{m+1} . Each application of $\mathcal{D}$ raises the outer index by one according to Equation (70). After $n - 1$ applications, the final index is $m + n$. Setting $m = N - n$ and recording the ordered operators gives Equation (73). Since $n \leq N$, only finitely many perturbation orders contribute to a fixed block. $\square$

6.3 The first four inverse blocks

For reference, the next coefficient after Equation (8)–Equation (10) is

$$\begin{aligned} a_4(\lambda) = & -\frac{5}{2654208} + \frac{6300C + 5629}{34836480\lambda} - \frac{37800C^2 + 12440C + 63}{5806080\lambda^2} \\ & + \frac{3024000C^3 + 415800C^2 - 12440C - 63}{29030400\lambda^3} \\ & - \frac{C^2(1800C^2 - 220C - 33)}{2880\lambda^4} - \frac{C^2(1360C^2 - 20C - 3)}{960\lambda^5} \\ & - \frac{35C^4}{24\lambda^6} - \frac{5C^4}{8\lambda^7}. \end{aligned} \quad (74)$$

The constant terms

$$\frac{1}{24}, \quad -\frac{1}{1152}, \quad \frac{1}{27648}, \quad -\frac{5}{2654208}, \dots \quad (75)$$

are precisely the coefficients of $\sqrt{1 + q/12} - 1$, as predicted by Equation (57).

A compact alternative notation is

$$\eta := \frac{\lambda}{24} - C. \quad (76)$$

Then

$$a_1 = \frac{\eta}{\lambda}, \quad (77)$$

$$a_2 = -\frac{\eta^2}{2\lambda^3} - \frac{\eta(12\eta - 1)}{24\lambda^2} + \frac{7}{5760\lambda}, \quad (78)$$

$$a_3 = \frac{\eta^3}{2\lambda^5} + \frac{\eta^2(40\eta - 3)}{48\lambda^4} + \frac{\eta(960\eta^2 - 160\eta + 1)}{1920\lambda^3} - \frac{7(72\eta - 1)}{138240\lambda^2} - \frac{31}{161280\lambda}. \quad (79)$$

This version makes the direct contributions of κ_1 and κ_2 more visible.

7 Analytic validity, remainder transfer, and certification

Formal reversion becomes an asymptotic theorem because the source expansion is uniform and the derivative of the dominant phase is large.

7.1 Residual of a formal truncation

For $N \geq 0$, define

$$t_N := T + \sum_{n=1}^N a_n(\lambda) T^{1-2n}, \quad z_N := \frac{1}{2} + t_N. \quad (80)$$

Lemma 7.1 (Residual order). *For every fixed $N \geq 0$,*

$$\log K(z_N) - Y = O_N(\lambda T^{-2N}) \quad (Y \rightarrow +\infty). \quad (81)$$

Proof. Divide the source equation by T^2 as in Equation (46). By construction, the coefficients of $q, q^2, \dots, q^N$ vanish after substituting the truncated series for u . The first possible uncanceled normal-ized term is of order λq^{N+1} ; the factor λ is present because the linear part of Ψ_λ is λu . Multiplication by $T^2 = q^{-1}$ gives $O(\lambda T^{-2N})$. The source remainder $O(t^{-2N})$ in Equation (29) has the same or smaller order. $\square$

7.2 Transfer from residual to inverse error

The centered derivative satisfies

$$\mathcal{L}'(t) = t \log t + O(t^{-1}) \quad (82)$$

uniformly on the positive real axis. Since $t_N/T \rightarrow 1$,

$$\mathcal{L}'(t) = T\lambda(1 + o(1)) \quad (83)$$

uniformly between t_N and the exact root for all sufficiently large Y .

Theorem 7.2 (Real-axis asymptotic inverse). *Let $t_* := K_+^{-1}(y) - \frac{1}{2}$. For every fixed $N \geq 0$,*

$$t_* - t_N = O_N(T^{-2N-1}). \quad (84)$$

Equivalently, Equation (11) holds.

Proof. By the mean value theorem, for some point ξ between t_N and t_* ,

$$t_N - t_* = \frac{\mathcal{L}(t_N) - \mathcal{L}(t_*)}{\mathcal{L}'(\xi)} = \frac{\log K(z_N) - Y}{\mathcal{L}'(\xi)}. \quad (85)$$

Use Equations (81) and (83) to obtain $O(\lambda T^{-2N})/(T\lambda) = O(T^{-2N-1})$. $\square$

Remark 7.3 (How many forward coefficients are required?). To construct $a_1, \dots, a_N$, it is enough to know the centered forward series through $\kappa_{N-1} t^{-2N+2}$. The next source term contributes at the same scale as the first omitted inverse block and therefore belongs in the remainder.

7.3 A posteriori residual certificates

Asymptotic order does not by itself certify a finite numerical computation. Monotonicity supplies a direct certificate.

Proposition 7.4 (Residual-to-error bound). *Let $Y > 0$, let $z_* = K_+^{-1}(e^Y)$, and suppose an interval $I = [a, b] \subset [2, \infty)$ contains both z_* and an approximation $\hat{z}$. Then*

$$|\hat{z} - z_*| \leq \frac{|\log K(\hat{z}) - Y|}{\inf_{x \in I} \left(x - \frac{1}{2} - \frac{1}{2} \log(2\pi) + \log \Gamma(x) \right)}. \quad (86)$$

The denominator is positive. Since the derivative is increasing on $[2, \infty)$, it may be replaced by its value at a .

Proof. Apply the mean value theorem to $\log K$ and use Equation (19) and Theorem 2.1. $\square$

A first-order residual correction is

$$z_* = \hat{z} - \frac{\log K(\hat{z}) - Y}{\hat{z} - \frac{1}{2} - \frac{1}{2} \log(2\pi) + \log \Gamma(\hat{z})} + O\left(\frac{R^2 \sup_I |1 + \psi|}{(\inf_I (\log K)')^3}\right), \quad (87)$$

where $R = \log K(\hat{z}) - Y$. Thus the transseries is also an excellent starting value for Newton or Halley iteration.

Scale and branch warning

All asymptotic series in this article are Poincaré expansions. The Bernoulli coefficients grow factorially, so summing indefinitely at fixed y is not justified. For high numerical accuracy, truncate near the least term or use a small fixed number of inverse blocks followed by a certified Newton step.

8 Rank-two structure and logarithmic re-expansion

The Lambert anchor packages an infinite logarithmic expansion into the single variable w . This section unpacks the structure and also gives a second resummation that reverses the order of the two scales.

8.1 The ordinary large-argument expansion of W

Let

$$L_1 = \log\left(\frac{4Y}{e}\right), \quad L_2 = \log L_1. \quad (88)$$

The principal Lambert function has the classical expansion [4, 5]

$$\begin{aligned} w = L_1 - L_2 + \frac{L_2}{L_1} + \frac{L_2(-2 + L_2)}{2L_1^2} \\ + \frac{L_2(6 - 9L_2 + 2L_2^2)}{6L_1^3} + O\left(\frac{L_2^4}{L_1^4}\right). \end{aligned} \quad (89)$$

Consequently,

$$\begin{aligned} T = 2\sqrt{\frac{Y}{L_1}} \left[1 + \frac{L_2}{2L_1} + \frac{L_2(3L_2 - 4)}{8L_1^2} \right. \\ + \frac{L_2(5L_2^2 - 16L_2 + 8)}{16L_1^3} \\ \left. + \frac{L_2(105L_2^3 - 568L_2^2 + 720L_2 - 192)}{384L_1^4} + O\left(\frac{L_2^5}{L_1^5}\right) \right]. \end{aligned} \quad (90)$$

Every finite truncation of Equation (90) is still much larger than the first outer correction a_1/T . Indeed,

$$\frac{a_1/T}{T} = O\left(\frac{L_1}{Y}\right) = o(L_1^{-M}) \quad (M \geq 0). \quad (91)$$

8.2 The fully expanded nested transseries

Combining Equations (7) and (89) gives a conventional iterated-logarithm transseries of the shape

$$K_+^{-1}(y) \sim \frac{1}{2} + \sum_{n \geq 0} 2^{1-2n} \left(\frac{L_1}{Y}\right)^{n-1/2} \sum_{r \geq 0} \frac{P_{n,r}(L_2)}{L_1^r}, \quad (92)$$

where each $P_{n,r}$ is a polynomial. The convention for $n = 0$ is $(L_1/Y)^{-1/2} = \sqrt{Y/L_1}$, and $P_{0,0} = 1$. The first row of these polynomials is displayed explicitly in Equation (90). All later rows follow mechanically by substituting Equation (89) into $a_n((w+1)/2)T^{1-2n}$.

Scale and branch warning

The double sum in Equation (92) must be truncated according to the transseries ordering, not by a naive rectangle. For every fixed r , all terms in the leading $n = 0$ row dominate the entire $n = 1$ row. The W -anchored form automatically respects this hierarchy and is therefore the recommended computational representation.

8.3 Reversing the hierarchy: $1/\lambda$ first

Equation (54) permits the alternative expansion

$$u(q, \lambda) \sim U_0(q) + \frac{U_1(q)}{\lambda} + \frac{U_2(q)}{\lambda^2} + \dots \quad (93)$$

This resums infinitely many outer blocks at each fixed slow order.

Set

$$r = r(q) := \sqrt{1 + \frac{q}{12}}. \quad (94)$$

Then the leading function is exactly

$$U_0(q) = r - 1. \quad (95)$$

The next one is also explicit.

Proposition 8.1 (First slow correction). *With the source series interpreted formally in q ,*

$$U_1(q) = -\frac{1}{r} \left\{ \frac{1}{2} \log r + \left(12C - \frac{1}{4}\right) (r^2 - 1) + \sum_{m \geq 1} \kappa_m q^{m+1} r^{-2m} \right\}. \quad (96)$$

Proof. Insert $u = U_0 + \lambda^{-1}U_1 + \dots$ into Equation (54). At order λ^0 , $A(U_0) = q/24$, giving Equation (95). At order λ^{-1} ,

$$A'(U_0)U_1 + R(U_0, q) = 0. \quad (97)$$

Since $A'(U_0) = 1 + U_0 = r$, it remains to simplify $R(r-1, q)$. Using $q = 12(r^2 - 1)$, one finds

$$B(r-1) + q \left(C - \frac{\log r}{24} \right) = \frac{1}{2} \log r + \left(12C - \frac{1}{4}\right) (r^2 - 1), \quad (98)$$

which proves the formula. $\square$

All higher U_j satisfy a triangular functional recurrence. For $j \geq 1$, let

$$V_{j-1}(s) := U_0 + \sum_{i=1}^{j-1} s^i U_i. \quad (99)$$

Then

$$rU_j + \frac{1}{2} \sum_{i=1}^{j-1} U_i U_{j-i} + [s^{j-1}] R(V_{j-1}(s), q) = 0. \quad (100)$$

This representation is useful when many outer powers of q are needed but only a few inverse powers of λ are desired.

Remark 8.2 (Two valid but different summation directions). The expansion $u = \sum_n a_n(\lambda) q^n$ is outer-first and is asymptotically canonical because $q \ll \lambda^{-M}$. The expansion $u = \sum_j \lambda^{-j} U_j(q)$ is a formal resummation across infinitely many outer blocks. It is algebraically useful, but analytic claims about the q -series U_j require separate convergence or summability analysis.

9 An accelerated Lambert anchor

The leading Lambert anchor treats the constant and logarithmic terms as a perturbation. A different elementary model absorbs both of them exactly and also captures a nontrivial infinite string of inverse powers. This gives a markedly sharper closed approximation before any transseries correction is added.

Set

$$d := \frac{1}{12}, \quad \widehat{\Phi}(t) := \frac{1}{4}(t^2 - d)(\log(t^2 - d) - 1) + C. \quad (101)$$

For positive large t , expansion of the logarithm gives

$$\widehat{\Phi}(t) = \Phi(t) + C - \frac{1}{24} \log t + \sum_{m \geq 1} \widehat{\kappa}_m t^{-2m}, \quad \widehat{\kappa}_m := \frac{d^{m+1}}{4m(m+1)}. \quad (102)$$

Indeed,

$$(1-x) \log(1-x) = -x + \sum_{j \geq 2} \frac{x^j}{j(j-1)}, \quad (103)$$

with $x = d/t^2$, and the constant terms cancel. Therefore the exact centered logarithm may be written

$$\mathcal{L}(t) \sim \widehat{\Phi}(t) + \widehat{H}(t), \quad \widehat{H}(t) := \sum_{m \geq 1} \rho_m t^{-2m}, \quad (104)$$

where

$$\rho_m := \kappa_m - \widehat{\kappa}_m. \quad (105)$$

The first residual coefficients are

$$\rho_1 = -\frac{1}{480}, \quad \rho_2 = \frac{61}{362880}, \quad \rho_3 = -\frac{433}{4354560}. \quad (106)$$

9.1 Exact inversion of the accelerated model

Let $Y = \log y$, and assume $Y > C$. Define

$$\widehat{w} := W_0\left(\frac{4(Y-C)}{e}\right), \quad S := \frac{4(Y-C)}{\widehat{w}}, \quad \widehat{T} := \sqrt{S+d}. \quad (107)$$

Then

$$\widehat{\Phi}(\widehat{T}) = Y. \quad (108)$$

To see this, set $X = \widehat{T}^2 - d = S$. The defining equation becomes

$$X(\log X - 1) = 4(Y - C), \quad (109)$$

and the substitution $\widehat{w} = \log X - 1$ reduces it to $\widehat{w}e^{\widehat{w}} = 4(Y - C)/e$.

Accelerated one-line inverse

For $y \rightarrow +\infty$, with $Y = \log y$,

$$K_+^{-1}(y) = \frac{1}{2} + \sqrt{\frac{4(Y - C)}{W_0(4(Y - C)/e)}} + \frac{1}{12} + O\left(\frac{1}{T^3 \log T}\right). \quad (110)$$

Here $C = \log A - \frac{1}{8} = -\zeta'(-1) - \frac{1}{24}$, and either T or $\widehat{T}$ may be used inside the displayed error term.

The gain over the uncorrected anchor $z \approx \frac{1}{2} + T$ is two outer powers: the latter has error $O(T^{-1})$, whereas Equation (110) has error $O(T^{-3}/\log T)$. The acceleration is not accidental: it resums the complete λ^0 row $U_0(q) = \sqrt{1 + q/12} - 1$, shifts the constant C into the Lambert argument, and incorporates the logarithmic correction into the modified quadratic variable $t^2 - d$.

9.2 All-orders corrections around the accelerated anchor

Since

$$\widehat{\Phi}'(t) = \frac{t}{2} \log(t^2 - d), \quad (111)$$

define the anchor derivative operator

$$\widehat{\mathcal{D}} := \frac{2}{t \log(t^2 - d)} \frac{d}{dt}. \quad (112)$$

The perturbative formula of Theorem 6.1 gives the formal accelerated inverse

$$t \sim \widehat{T} + \sum_{n \geq 1} \frac{(-1)^n}{n!} \widehat{\mathcal{D}}^{n-1} \left(\frac{\widehat{H}(t)^n}{\widehat{\Phi}'(t)} \right) \Big|_{t=\widehat{T}}. \quad (113)$$

Every fixed inverse order receives contributions from only finitely many perturbation orders. The first correction follows solely from $\rho_1 = -1/480$:

$$t = \widehat{T} + \frac{1}{240\widehat{T}^3 \log(\widehat{T}^2 - 1/12)} + O\left(\frac{1}{\widehat{T}^5 \log \widehat{T}} + \frac{1}{\widehat{T}^7 (\log \widehat{T})^3}\right). \quad (114)$$

The two terms displayed in the remainder come respectively from the next source coefficient and from nonlinear reversion of the first one.

Proposition 9.1 (Accelerated error transfer). *For $N \geq 0$, let $\widehat{t}_N$ denote the truncation of Equation (113) selected so that substitution into the accelerated forward transseries cancels every residual term through source order t^{-2N} ; thus $\widehat{t}_0 = \widehat{T}$. Then, on the positive real axis,*

$$K_+^{-1}(e^Y) - \frac{1}{2} - \widehat{t}_N = O_N\left(\frac{\widehat{T}^{-2N-3}}{\log \widehat{T}}\right), \quad Y \rightarrow +\infty. \quad (115)$$

For $N = 0$, this is Equation (110). For $N = 1$, retaining the correction in Equation (114) cancels $\rho_1 t^{-2}$ and moves the leading error to the scale stated inside that formula's remainder.

Proof. By construction, the accelerated forward residual is $O(t^{-2N-2})$. Division by $\widehat{\Phi}'(t) \asymp t \log t$, followed by the same mean-value argument as in [Theorem 7.2](#), gives $O(t^{-2N-3}/\log t)$. The retained formal reversion terms cancel all larger nonlinear contributions, and $t \sim \widehat{T}$ converts this to [Equation \(115\)](#). $\square$

Remark 9.2 (Which anchor should be used?). The basic anchor is structurally simpler and exposes the polynomial block coefficients $a_n(\lambda)$. The accelerated anchor is usually superior for a one-line numerical approximation. They encode the same inverse germ; they merely choose different portions of the forward transseries as the exactly invertible model.

10 Complex sectorial inverse germs

The positive real inverse uses the principal logarithm and W_0 , but the same construction describes complex inverse germs. Because both the logarithm of the target and Lambert W are multivalued, two branch indices appear naturally.

Fix a branch of $\log K(t + \frac{1}{2})$ in a sector

$$|\arg t| \leq \pi - \delta, \quad \delta > 0, \quad (116)$$

where the centered Stirling expansion is uniform. For a nonzero target y , choose

$$Y_\ell := \text{Log } y + 2\pi i \ell, \quad \ell \in \mathbb{Z}, \quad (117)$$

and a branch W_k of Lambert W . Set

$$w_{k,\ell} := W_k\left(\frac{4Y_\ell}{e}\right), \quad \lambda_{k,\ell} := \frac{w_{k,\ell} + 1}{2}, \quad T_{k,\ell} := \exp(\lambda_{k,\ell}). \quad (118)$$

Then $\Phi(T_{k,\ell}) = Y_\ell$ on the corresponding logarithmic sheet.

Theorem 10.1 (Sectorial complex inverse transseries). *Let a pair (k, ℓ) vary along a path such that*

$$|T_{k,\ell}| \longrightarrow \infty, \quad |\arg T_{k,\ell}| \leq \pi - \delta, \quad |w_{k,\ell} + 1| \geq \varepsilon > 0. \quad (119)$$

Assume also that the chosen local solution avoids critical points of $\log K$. Then there is a sectorial inverse germ with

$$z_{k,\ell}(y) \sim \frac{1}{2} + T_{k,\ell} + \sum_{n \geq 1} a_n(\lambda_{k,\ell}) T_{k,\ell}^{1-2n}, \quad (120)$$

where the same rational functions a_n as on the real branch occur. For every fixed N , the error is

$$O_{N,\delta,\varepsilon}(|T_{k,\ell}|^{-2N-1}) \quad (121)$$

through any closed subsector on which the derivative remains uniformly nonzero.

Proof. The centered forward expansion and its differentiated form are uniform in closed subsectors of [Equation \(116\)](#); this follows from the corresponding sectorial Stirling expansions for Γ and Barnes G [5, §5.11, §5.17]. The formal coefficient calculation is algebraic and is unchanged after complexification. Condition $|w+1| \geq \varepsilon$ excludes the critical value of the Lambert anchor, for which $\Phi'(T) = T(w+1)/2$ vanishes. The analytic inverse-function theorem and the complex mean-value replacement based on contour estimates then transfer the residual order to the inverse order. $\square$

Scale and branch warning

The indices (k, ℓ) label asymptotic constructions, not necessarily globally distinct single-valued branches of an entire inverse. Analytic continuation can permute germs, two index pairs can describe the same germ on overlapping sheets, and singularities occur at critical values of K as well as at the Lambert branch point $w = -1$. Formula Equation (120) is therefore a sectorial local statement.

The accelerated anchor also complexifies. Replacing Y by Y_ℓ , choosing a branch $W_k(4(Y_\ell - C)/e)$, and choosing the compatible square root gives

$$\widehat{T}_{k,\ell}^2 = \frac{4(Y_\ell - C)}{W_k(4(Y_\ell - C)/e)} + \frac{1}{12}. \quad (122)$$

The operator formula Equation (113) then applies on any domain where $\log(\widehat{T}^2 - 1/12)$ and the square root are consistently chosen and the anchor derivative does not vanish.

11 A general theory for Lambert-invertible logarithmic transseries

The calculation above is an instance of a broad and stable inversion mechanism. Its essential feature is a dominant monomial multiplied by one logarithm. Lambert W inverts that dominant phase exactly, while a differential operator converts every lower-order source monomial into a strictly smaller inverse monomial.

11.1 The exact generalized Lambert anchor

Let

$$F_0(x) := ax^p(\log x - b), \quad a \neq 0, \quad p > 0, \quad (123)$$

and solve $F_0(X) = Y$. Put

$$w := W_k\left(\frac{p}{a}e^{-pb}Y\right), \quad L := b + \frac{w}{p}, \quad X := e^L = \left(\frac{pY}{aw}\right)^{1/p}. \quad (124)$$

Then $F_0(X) = Y$. The logarithmic slope variable is

$$\Lambda := p(L - b) + 1 = w + 1, \quad (125)$$

and

$$F'_0(X) = aX^{p-1}\Lambda. \quad (126)$$

Thus the only intrinsic degeneracy of the anchor occurs at the Lambert critical point $w = -1$.

11.2 A one-grid source class

Fix $\sigma > 0$ and define the outer monomial

$$Q := X^{-\sigma}. \quad (127)$$

Consider a source transseries of the form

$$F(x) = F_0(x) + H(x), \quad H(x) = x^{p-\sigma}\mathcal{H}(x^{-\sigma}, \log x), \quad (128)$$

where

$$\mathcal{H}(Q, L) = \sum_{m \geq 0} h_m(L)Q^m. \quad (129)$$

The coefficients h_m belong to a differential algebra $\mathcal{A}$ that is closed under d/dL , multiplication, and division by $a\Lambda$. Typical choices are rational functions in L , Laurent polynomials in Λ , or logarithmic transseries in slower scales.

Definition 11.1 (Lambert-grid inverse transseries). A formal inverse associated with Equation (128) is an expansion

$$x = X \left(1 + \sum_{N \geq 1} c_N(L) Q^N \right), \quad c_N \in \mathcal{A}[\Lambda^{-1}], \quad (130)$$

ordered first by the powers of Q and then by the chosen asymptotic order inside the coefficient algebra.

The anchor derivative operator is

$$\mathcal{D}_0 := \frac{1}{F'_0(X)} \frac{d}{dX}. \quad (131)$$

For every exponent ρ and coefficient function $f(L)$,

$$\mathcal{D}_0(X^\rho f(L)) = X^{\rho-p} \mathcal{B}_\rho f(L), \quad \mathcal{B}_\rho f := \frac{f'(L) + \rho f(L)}{a\Lambda}. \quad (132)$$

Each application lowers the power of X by exactly p , while remaining inside the coefficient differential algebra.

Theorem 11.2 (Universal inverse formula). *For the source Equation (128), the formal inverse satisfying $F(x) = Y = F_0(X)$ is*

$$x \sim X + \sum_{n \geq 1} \frac{(-1)^n}{n!} \mathcal{D}_0^{n-1} \left(\frac{H(X)^n}{F'_0(X)} \right). \quad (133)$$

At each prescribed power of Q , only finitely many values of n contribute. Consequently Equation (133) is well defined as a formal transseries independently of convergence of the perturbation series in an auxiliary parameter.

Proof. Formula Equation (133) is Equation (62) with the auxiliary parameter set to one. A monomial from H^n/F'_0 has initial size

$$X^{n(p-\sigma)-(p-1)-m\sigma} = X^{(n-1)p-n\sigma+1-m\sigma}, \quad (134)$$

where $m \geq 0$ is its total source-grid index. After $n-1$ applications of $\mathcal{D}_0$, Equation (132) changes this to

$$X^{1-(m+n)\sigma} = XQ^{m+n}. \quad (135)$$

Hence the coefficient of XQ^N can only receive contributions from $1 \leq n \leq N$, with $m = N - n$. This proves local finiteness and the claimed shape. $\square$

Theorem 11.3 (Ordered coefficient formula). *Let*

$$\rho_{n,N} := (n-1)p - N\sigma + 1. \quad (136)$$

Then the coefficient in Equation (130) is

$$c_N(L) = \sum_{n=1}^N \frac{(-1)^n}{n!} (\mathcal{B}_{\rho_{n,N}-(n-2)p} \circ \cdots \circ \mathcal{B}_{\rho_{n,N}-p} \circ \mathcal{B}_{\rho_{n,N}}) \times \left(\frac{[Q^{N-n}] \mathcal{H}(Q, L)^n}{a\Lambda} \right). \quad (137)$$

For $n = 1$, the operator composition is empty. For $n \geq 2$, it contains exactly $n-1$ factors, and the rightmost factor acts first.

Proof. The coefficient $[Q^{N-n}] \mathcal{H}^n$ in H^n/F'_0 is multiplied by $X^{\rho_{n,N}}$. Successive applications of $\mathcal{D}_0$ lower its power by p , so Equation (132) produces the displayed ordered list of $\mathcal{B}$ -operators. The final power is $X^{1-N\sigma}$, as required. $\square$

11.3 Normalized implicit recurrence

An alternative construction avoids the perturbative operator formula. Set

$$x = X(1 + u), \quad u = \sum_{N \geq 1} c_N Q^N. \quad (138)$$

After division by aX^p , the equation $F(x) = F_0(X)$ becomes

$$0 = (1 + u)^p (L + \log(1 + u) - b) - (L - b) + \frac{Q}{a} (1 + u)^{p-\sigma} \mathcal{H}(Q(1 + u)^{-\sigma}, L + \log(1 + u)). \quad (139)$$

The derivative of its first line with respect to u at $u = 0$ is Λ . Therefore the coefficient of c_N in the Q^N -equation is Λ , and the recurrence is triangular:

$$c_N = -\frac{1}{\Lambda} [Q^N] \mathcal{F} \left(\sum_{j=1}^{N-1} c_j Q^j, Q; L \right), \quad (140)$$

where $\mathcal{F}$ denotes the left-hand side of Equation (139). Formula Equation (140) is often the simplest route for computer algebra; Equation (137) is usually the clearest theoretical formula.

11.4 Asymptotic validity

Theorem 11.4 (Residual-to-inverse transfer for the general class). *Suppose F is analytic in a real ray or closed complex sector, the source expansion is differentiable term by term there, and*

$$H(x) = x^{p-\sigma} \sum_{m=0}^{N-1} h_m(\log x) x^{-m\sigma} + O(x^{p-(N+1)\sigma} \mathcal{A}_N(x)), \quad (141)$$

where $\mathcal{A}_N$ is a controlled slow factor. Assume

$$|\Lambda| \geq \varepsilon > 0, \quad |F'(x)| \asymp |a\Lambda X^{p-1}| \quad (142)$$

uniformly near the desired root. If x_N is the inverse transseries truncated after $c_N X Q^N$, then

$$x - x_N = O\left(X^{1-(N+1)\sigma} \tilde{\mathcal{A}}_N(X)\right), \quad (143)$$

where $\tilde{\mathcal{A}}_N$ differs from $\mathcal{A}_N$ only by factors generated inside the coefficient differential algebra, in particular by powers of Λ^{-1} .

Proof. Substitution of the recursively determined coefficients cancels the normalized residual through Q^N . Restoring the factor aX^p leaves a residual of order $X^{p-(N+1)\sigma}$ times a slow factor. Division by the local slope $F'(x) \asymp a\Lambda X^{p-1}$, using either the real mean-value theorem or a sectorial inverse-function estimate, produces $X^{1-(N+1)\sigma}$ and the indicated slow factors. $\square$

11.5 Beyond a single grid

The one-grid hypothesis is convenient, not essential. Suppose instead that

$$H(x) = \sum_{\gamma \in \Gamma} x^{p-\gamma} h_\gamma(\log x), \quad (144)$$

where $\Gamma \subset (0, \infty)$ is well ordered and locally finite. Products of source monomials generate the additive semigroup

$$\langle \Gamma \rangle := \{\gamma_1 + \cdots + \gamma_n : n \geq 1, \gamma_j \in \Gamma\}. \quad (145)$$

The same power count as in Equation (135) shows that the relative inverse support is contained in $\langle \Gamma \rangle$. Whenever that semigroup is locally finite—as it is for every finitely generated positive grid—the perturbative formula is coefficientwise finite. Hahn or logarithmic transseries coefficient fields may be used for the slow coefficients. This is the natural setting for mixtures of integer powers, fractional powers, iterated logarithms, and exponentially small sectors; see [8, 1] for the broader transseries framework.

General Lambert-grid inversion recipe

Given a large equation $F(x) = Y$ whose dominant part is $ax^p(\log x - b)$:

1. invert the dominant phase exactly by Equation (124);
2. choose an outer support for the lower-order source and verify local finiteness under addition;
3. compute coefficients either by the normalized recurrence Equation (140) or by the ordered operator formula Equation (137);
4. retain Lambert W in the final expression unless a fully logarithmic expansion is explicitly needed;
5. certify a finite approximation from its residual and a lower bound on $|F'|$.

12 Examples and neighboring inverse problems

The parameters in the general theory clarify both what is special about the K -function and what is universal. The following table records the dominant phase, the first outer spacing, and the exact Lambert anchor in several standard problems. In each row Y denotes the logarithm of the original large target when the special function itself, rather than its logarithm, is being inverted.

Table 1: Lambert anchors for related logarithmic inverse problems.

Problem	(a, p, b)	σ	Dominant anchor X
$\log \Gamma(x) = Y$	$(1, 1, 1)$	1	$Y/W(Y/e)$
Centered	$(\frac{1}{2}, 2, \frac{1}{2})$	2	$2\sqrt{Y/W(4Y/e)}$
$\log K(t + \frac{1}{2}) = Y$	$(\frac{1}{2}, 2, \frac{3}{2})$	1	$2\sqrt{Y/W(4Y/e^3)}$
$\log G(x + 1) = Y$	$(\frac{1}{2}, 2, \frac{3}{2})$	1	$(p^2 Y/W(p^2 Y/e))^{1/p}$
Order- r hyperfactorial, $p = r + 1$	$(1/p, p, 1/p)$	typically 1	

12.1 The inverse gamma function

Stirling's expansion has the form

$$\log \Gamma(x) \sim x(\log x - 1) - \frac{1}{2} \log x + \frac{1}{2} \log(2\pi) + \sum_{m \geq 1} \frac{B_{2m}}{2m(2m-1)x^{2m-1}}. \quad (146)$$

Thus $a = 1$, $p = 1$, $b = 1$, and the correction begins at x^0 , so $\sigma = 1$. With

$$X = \frac{Y}{W(Y/e)}, \quad \Lambda = W(Y/e) + 1, \quad (147)$$

the first general coefficient is

$$c_1 = -\frac{h_0(L)}{\Lambda}, \quad h_0(L) = -\frac{L}{2} + \frac{1}{2} \log(2\pi). \quad (148)$$

This recovers the familiar Lambert description of the large inverse gamma branch while placing all subsequent corrections in a single automatic calculus.

12.2 The centered inverse K -function

For $x = t$, the centered source has

$$a = \frac{1}{2}, \quad p = 2, \quad b = \frac{1}{2}, \quad \sigma = 2, \quad h_0(L) = C - \frac{L}{24}. \quad (149)$$

Since $a\Lambda = \frac{1}{2}(w+1) = \lambda$, the universal first coefficient

$$c_1 = -\frac{h_0(L)}{a\Lambda} \quad (150)$$

becomes

$$c_1 = \frac{1}{24} - \frac{C}{\lambda} = a_1(\lambda), \quad (151)$$

exactly as in Equation (8). The unusually large spacing $\sigma = 2$ is the benefit of the canonical half-shift and parity cancellation.

12.3 The inverse Barnes G -function

The standard Barnes expansion is

$$\begin{aligned} \log G(x+1) &\sim \frac{x^2}{2} \log x - \frac{3x^2}{4} + \frac{x}{2} \log(2\pi) - \frac{1}{12} \log x + \zeta'(-1) \\ &+ \sum_{m \geq 1} \frac{B_{2m+2}}{4m(m+1)x^{2m}}. \end{aligned} \quad (152)$$

The dominant phase has $(a, p, b) = (\frac{1}{2}, 2, \frac{3}{2})$, but the $x \log(2\pi)/2$ correction begins one power below the dominant term. Consequently $\sigma = 1$, and the inverse contains every integer outer power rather than only the even lattice obtained for the centered K -function. This comparison explains why the identity

$$K(z)G(z) = \Gamma(z)^{z-1} \quad (153)$$

does not by itself reveal the sparse inverse support of K : the sparsity appears only after cancellation and centering.

12.4 Higher generalized hyperfactorials

For an integer $r \geq 0$, consider the discrete products

$$H_r(n) := \prod_{k=1}^n k^{k^r}. \quad (154)$$

Euler–Maclaurin summation gives, with $p = r + 1$,

$$\log H_r(n) = \frac{n^p}{p} \log n - \frac{n^p}{p^2} + O(n^{p-1} \log n). \quad (155)$$

Any normalized continuous interpolation having this asymptotic form therefore has the dominant parameters

$$a = \frac{1}{p}, \quad b = \frac{1}{p}, \quad X = \left(\frac{p^2 Y}{W(p^2 Y/e)} \right)^{1/p}. \quad (156)$$

The lower Euler–Maclaurin terms normally begin at relative order X^{-1} , so the generic grid spacing is $\sigma = 1$. Special shifts can eliminate some sublattices, just as the half-shift does for K . Determining such a shift amounts to choosing a center at which the relevant Bernoulli polynomials vanish.

12.5 The first two universal coefficients

For later reuse, write the source as in [Equation \(129\)](#). The first two relative inverse coefficients obtained from [Equation \(137\)](#) are

$$c_1 = -\frac{h_0}{a\Lambda}, \quad (157)$$

$$c_2 = -\frac{h_1}{a\Lambda} + \frac{1}{2}\mathcal{B}_{p-2\sigma+1}\left(\frac{h_0^2}{a\Lambda}\right). \quad (158)$$

These formulas already include differentiation of all logarithmic coefficient functions through $\mathcal{B}_\rho$. They are therefore valid without assuming that the h_m are constants.

13 Symbolic generation and stable numerical evaluation

The recurrence and operator formulas are straightforward to automate. This section gives reproducible pseudocode in Python/SymPy notation and a stable workflow for evaluating the inverse without forming the astronomically large number $y = e^Y$.

13.1 Generating the centered source coefficients

The Bernoulli-polynomial formula [Equation \(28\)](#) is directly implementable:

```
import sympy as sp

m = sp.symbols("m", integer=True, positive=True)

def kappa(j: int) -> sp.Expr:
    if j < 1:
        raise ValueError("j must be positive")
    return sp.factor(
        -sp.bernoulli(2*j + 2, sp.Rational(1, 2))
        / (4*j*(j + 1)*(2*j + 1))
    )

print([kappa(j) for j in range(1, 6)])
# [-7/5760, 31/161280, -127/1290240,
#  511/4866048, -1414477/7380172800]
```

13.2 Triangular generation of inverse blocks

The following routine implements [Equation \(49\)](#). Every expression is truncated before coefficient extraction, preventing irrelevant high powers from expanding.

```
import sympy as sp

q, lam, C = sp.symbols("q lam C")

def trunc(expr: sp.Expr, order: int) -> sp.Expr:
    return sp.series(expr, q, 0, order + 1).removeO().expand()

def inverse_blocks(N: int) -> list[sp.Expr]:
    if N < 1:
        return []

    u = sp.Integer(0)
```

```

blocks: list[sp.Expr] = []

for n in range(1, N + 1):
    log1pu = trunc(sp.log(1 + u), n)
    psi = sp.Rational(1, 2)*(1 + u)**2 * (
        lam + log1pu - sp.Rational(1, 2)
    ) - sp.Rational(1, 2)*(lam - sp.Rational(1, 2))

    source = q*(C - (lam + log1pu)/24)
    for j in range(1, n):
        source += kappa(j)*q**(j + 1) * trunc(
            (1 + u)**(-2*j), n
        )

    residual_n = trunc(psi + source, n).coeff(q, n)
    a_n = sp.factor(-residual_n/lam)
    blocks.append(a_n)
    u = trunc(u + a_n*q**n, n)

return blocks

```

Because the coefficient equation is triangular, no nonlinear equation solver is invoked. Exact rational arithmetic is preserved, and C may remain a symbol until numerical evaluation.

13.3 Evaluation from $Y = \log y$

For very large targets, passing y itself to a floating-point routine is unnecessary and may overflow. The natural input is Y . The basic Lambert-anchored truncation is evaluated as follows.

```

from mpmath import mp

mp.dps = 80
Cnum = -mp.diff(mp.zeta, -1) - mp.mpf(1)/24

def inverse_K_from_logY(Y, coeff_functions):
    Y = mp.mpf(Y)
    w = mp.lambertw(4*Y/mp.e)
    lam = (w + 1)/2
    T = mp.e**lam

    z = mp.mpf(1)/2 + T
    for n, a_n in enumerate(coeff_functions, start=1):
        z += a_n(lam, Cnum) * T**(1 - 2*n)
    return z

```

A robust production implementation should also:

1. evaluate $T = 2\sqrt{Y/w}$ rather than e^λ when that is numerically better scaled;
2. increase working precision before subtracting a residual near zero;
3. use the exact derivative [Equation \(19\)](#) for a Newton correction;
4. certify the final answer with [Equation \(86\)](#) whenever a rigorous enclosure is needed.

Algorithm 13.1 (Lambert-transseries inverse of K). Given $Y = \log y$, a block order N , and a working precision:

1. Compute $w = W_0(4Y/e)$, $\lambda = (w + 1)/2$, and $T = 2\sqrt{Y/w}$.
2. Form $z_N = \frac{1}{2} + T + \sum_{n=1}^N a_n(\lambda)T^{1-2n}$.

3. Optionally replace z_N by one Newton iterate

$$z_N - \frac{(z_N - 1) \log \Gamma(z_N) - \log G(z_N) - Y}{z_N - \frac{1}{2} - \frac{1}{2} \log(2\pi) + \log \Gamma(z_N)}.$$

4. Estimate or bound the remaining error from the new residual.

The numerator in the Newton step uses the exact positive-real identity [Equation \(17\)](#); it avoids numerical integration in the defining formula for K .

13.4 Choosing a truncation order

For fixed N , the theorem gives an unambiguous asymptotic order. If many blocks are generated, however, the factorial growth of the Bernoulli source coefficients eventually propagates into the inverse. A practical adaptive rule is:

$$\text{stop before the first block whose magnitude exceeds its predecessor,} \quad (159)$$

then apply one Newton step. This is a numerical heuristic rather than a replacement for a remainder theorem. When only moderate precision is needed, three or four blocks are already far beyond the accuracy of the leading Lambert approximation over a wide range of Y .

14 High-precision numerical checks

The formulas were checked against the exact identity

$$\log K(z) = (z - 1) \log \Gamma(z) - \log G(z), \quad z > 0, \quad (160)$$

using high-precision arithmetic. The reference root was computed directly from $\log K(z) = Y$, and the reported numbers are absolute errors in z . No Newton correction was applied to the asymptotic values in the tables.

Table 2: Absolute error of the basic Lambert-anchored truncations $z_N = \frac{1}{2} + T + \sum_{n=1}^N a_n T^{1-2n}$.

Y	exact inverse z_*	$N = 0$	$N = 1$	$N = 2$	$N = 4$
10	4.966157945496709	9.162×10^{-3}	1.935×10^{-5}	1.347×10^{-7}	6.963×10^{-11}
100	10.915136726993085	1.070×10^{-3}	2.046×10^{-7}	5.165×10^{-10}	2.775×10^{-14}
1000	27.284589040600087	1.503×10^{-4}	2.134×10^{-8}	4.241×10^{-12}	4.537×10^{-18}
10^6	584.235864411692401	3.809×10^{-5}	2.524×10^{-13}	6.128×10^{-19}	2.420×10^{-30}

The $N = 3$ errors, omitted from the table for width, are respectively

$$6.036 \times 10^{-10}, \quad 2.690 \times 10^{-12}, \quad 3.032 \times 10^{-15}, \quad 8.319 \times 10^{-25}. \quad (161)$$

The nonmonotone comparison between $N = 3$ and $N = 4$ at $Y = 10$ is a normal finite-argument effect; asymptotic truncations are not required to improve monotonically at every point.

Several features are visible.

- The first basic correction removes roughly two to eight decimal orders over the displayed range, with the gain increasing rapidly with Y .
- Four basic blocks deliver about 14 correct decimal places already at $Y = 100$, and about 30 digits at $Y = 10^6$.
- The accelerated one-line anchor has nearly the accuracy of the first basic correction, while its first residual correction is competitive with two or three basic blocks.

These numerical observations agree with the proved orders; they are not used as evidence for the derivation.

Table 3: Absolute error of the accelerated anchor and of its first correction.

Y	anchor Equation (110)	anchor plus Equation (114)
10	1.559×10^{-5}	6.167×10^{-8}
100	7.865×10^{-7}	5.825×10^{-10}
1000	3.297×10^{-8}	3.706×10^{-12}
10^6	1.644×10^{-12}	3.894×10^{-19}

15 Conclusions

The inverse generalized hyperfactorial is governed by a sparse two-scale transseries once the correct variable is chosen. The central facts are:

1. The shift $t = z - \frac{1}{2}$ converts the logarithm of K into $\Phi(t) + C - \frac{1}{24} \log t$ plus only even inverse powers.
2. The dominant phase $\Phi(t) = \frac{1}{2}t^2 \log t - \frac{1}{4}t^2$ is inverted exactly by $w = W(4Y/e)$, not merely approximated by an iterated-logarithm series.
3. The inverse has the all-orders form $\frac{1}{2} + T + \sum_{n \geq 1} a_n(\lambda) T^{1-2n}$, with a triangular recurrence, an ordered operator formula, and a fixed-order remainder $O(T^{-2N-1})$.
4. Every K -coefficient block is a finite polynomial in λ^{-1} ; the constant row resums to $\sqrt{1 + q/12} - 1$.
5. A modified quadratic Lambert model absorbs the constant, logarithm, and an infinite rational tail, yielding a substantially sharper one-line approximation.
6. The same construction works for $ax^p(\log x - b)$ plus locally finite lower-order logarithmic transseries. Lambert inversion supplies the anchor, and Lagrange–Bürmann supplies a universal coefficient calculus.

The practical recommendation is equally simple: retain Lambert W as an exact structural component, use a small number of outer correction blocks, and finish with a residual-controlled Newton step when numerical accuracy is the goal. For formal work, the normalized recurrence is easiest to automate; for proofs and support analysis, the ordered differential-operator formula is the more informative representation.

A Formula sheet

This appendix collects the formulas needed for direct use without repeating the derivations.

A.1 Forward data

$$t = z - \frac{1}{2}, \quad \Phi(t) = \frac{1}{2}t^2 \log t - \frac{1}{4}t^2, \quad C = -\zeta'(-1) - \frac{1}{24}. \quad (162)$$

$$\log K\left(t + \frac{1}{2}\right) \sim \Phi(t) + C - \frac{1}{24} \log t + \sum_{m \geq 1} \kappa_m t^{-2m}. \quad (163)$$

$$\kappa_m = \frac{(1 - 2^{-2m-1})B_{2m+2}}{4m(m+1)(2m+1)}. \quad (164)$$

A.2 Basic Lambert anchor and inverse

$$Y = \log y, \quad w = W_0(4Y/e), \quad \lambda = \frac{w+1}{2}, \quad (165)$$

$$T = 2\sqrt{Y/w} = e^\lambda, \quad q = T^{-2} = \frac{w}{4Y}, \quad (166)$$

$$K_+^{-1}(y) \sim \frac{1}{2} + T + \sum_{n \geq 1} a_n(\lambda) T^{1-2n}. \quad (167)$$

The first block is

$$a_1 = \frac{1}{24} - \frac{C}{\lambda}. \quad (168)$$

The higher blocks through a_4 are given in [Equations \(9\), \(10\)](#) and [\(74\)](#). The normalized recurrence is

$$a_N = -\frac{1}{\lambda} [q^N] \mathcal{F}(U_{N-1}, q; \lambda), \quad U_{N-1} = \sum_{j=1}^{N-1} a_j q^j, \quad (169)$$

with $\mathcal{F}$ defined by [Equation \(46\)](#).

A.3 Accelerated anchor

$$\hat{w} = W_0\left(\frac{4(Y-C)}{e}\right), \quad \hat{T} = \sqrt{\frac{4(Y-C)}{\hat{w}} + \frac{1}{12}}, \quad (170)$$

$$K_+^{-1}(y) = \frac{1}{2} + \hat{T} + \frac{1}{240\hat{T}^3 \log(\hat{T}^2 - 1/12)} + \dots. \quad (171)$$

A.4 General Lambert-grid inversion

For

$$F(x) = ax^p(\log x - b) + x^{p-\sigma} \mathcal{H}(x^{-\sigma}, \log x), \quad (172)$$

put

$$w = W_k\left(\frac{p}{a} e^{-pb} Y\right), \quad X = \left(\frac{pY}{aw}\right)^{1/p}, \quad \Lambda = w + 1. \quad (173)$$

Then

$$x \sim X + \sum_{n \geq 1} \frac{(-1)^n}{n!} \left(\frac{1}{aX^{p-1}\Lambda} \frac{d}{dX} \right)^{n-1} \left(\frac{H(X)^n}{aX^{p-1}\Lambda} \right). \quad (174)$$

B Conventions and shift dictionary

Several neighboring conventions coexist in the literature. The formulas in this article use

$$K(x+1) = x^x K(x), \quad K(1) = 1, \quad K(n+1) = \prod_{k=1}^n k^k. \quad (175)$$

If the hyperfactorial is written

$$H(x) := K(x+1), \quad (176)$$

then

$$H_+^{-1}(y) = K_+^{-1}(y) - 1. \quad (177)$$

If a discrete source defines $H_n = \prod_{k=1}^{n-1} k^k$, then $H_n = K(n)$, whereas the convention $\prod_{k=1}^n k^k$ equals $K(n+1)$. These shifts do not change the Lambert scale, but they do change the constant term of an inverse expansion and therefore must be fixed before comparing formulas.

The Bernoulli convention is $B_1 = -1/2$, generated by

$$\frac{u}{e^u - 1} = \sum_{n \geq 0} B_n \frac{u^n}{n!}. \quad (178)$$

The symbol W_k denotes the k -th branch of Lambert W , characterized locally by $W_k(z)e^{W_k(z)} = z$. On the positive real axis all main formulas use W_0 .

C A complete high-precision verification script

The following compact script regenerates the coefficients by the triangular recurrence, computes the exact positive-real benchmark through Barnes G , and reproduces tables of inverse errors. It uses SymPy for exact symbolic algebra and mpmath for numerical evaluation.

```
import sympy as sp
from mpmath import mp

q, lam, Csym = sp.symbols("q lam C")

def kap(j):
    return sp.factor(
        -sp.bernoulli(2*j + 2, sp.Rational(1, 2))
        / (4*j*(j + 1)*(2*j + 1))
    )

def trunc(expr, n):
    return sp.series(expr, q, 0, n + 1).removeO().expand()

def make_blocks(N):
    u = sp.Integer(0)
    ans = []
    for n in range(1, N + 1):
        log1pu = trunc(sp.log(1 + u), n)
        psi = sp.Rational(1, 2)*(1 + u)**2 * (
            lam + log1pu - sp.Rational(1, 2)
        ) - sp.Rational(1, 2)*(lam - sp.Rational(1, 2))
        F = psi + q*(Csym - (lam + log1pu)/24)
        for j in range(1, n):
            F += kap(j)*q**(j + 1)*trunc((1 + u)**(-2*j), n)
        an = sp.factor(-trunc(F, n).coeff(q, n)/lam)
        ans.append(an)
        u = trunc(u + an*q**n, n)
    return ans

mp.dps = 100
Cnum = -mp.diff(mp.zeta, -1) - mp.mpf(1)/24
blocks = make_blocks(4)
block_num = [sp.lambdify((lam, Csym), a, "mpmath") for a in blocks]

def logK(z):
    return (z - 1)*mp.loggamma(z) - mp.log(mp.barnesg(z))

def exact_inverse(Y):
```

```

w = mp.lambertw(4*Y/mp.e)
guess = mp.mpf(1)/2 + 2*mp.sqrt(Y/w)
return mp.findroot(lambda z: logK(z) - Y, guess)

def basic_inverse(Y, N):
    w = mp.lambertw(4*Y/mp.e)
    la = (w + 1)/2
    T = 2*mp.sqrt(Y/w)
    z = mp.mpf(1)/2 + T
    for n in range(N):
        z += block_num[n](la, Cnum)*T**(-2*n - 1)
    return z

def accelerated_inverse(Y, corrected=False):
    wh = mp.lambertw(4*(Y - Cnum)/mp.e)
    Th = mp.sqrt(4*(Y - Cnum)/wh + mp.mpf(1)/12)
    z = mp.mpf(1)/2 + Th
    if corrected:
        z += 1/(240*Th**3*mp.log(Th**2 - mp.mpf(1)/12))
    return z

for Y in map(mp.mpf, [10, 100, 1000, 10**6]):
    root = exact_inverse(Y)
    print("Y =", Y, "root =", mp.nstr(root, 40))
    print("basic errors:",
          [mp.nstr(abs(basic_inverse(Y, N) - root), 12)
           for N in range(5)])
    print("accelerated errors:",
          [mp.nstr(abs(accelerated_inverse(Y, flag) - root), 12)
           for flag in (False, True)])

```

At positive real z , $G(z) > 0$ and the logarithms in the script are unambiguous. For complex tests, one must maintain compatible branches of $\log \Gamma$, $\log G$, $\log K$, and W along the continuation path.

D Project context and reproducibility

This note was prepared for the “Series and transseries” research-frontier folder of the ProveIt project:

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/series-and-transseries>.

The $\text{\LaTeX}$ source is standalone: it has no external figures, bibliography files, or input fragments. A standard `pdflatex` run performed twice is sufficient to resolve the table of contents and cross-references.

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